

NAVLE Differential Clusters

Companion to the Bread-and-Butter Study Guide. Each cluster below groups look-alike conditions that the NAVLE loves to test against each other — same presenting complaint, different underlying disease, distinguished by one or two key clues. Organized roughly by exam weight: canine/feline first, then equine, bovine, cross-species toxicology/regulatory topics, and a swine enteric disease recap.

Canine & Feline Clusters

1. Causes of PU/PD

Cause	Distinguishing Clue	Confirmatory Test
Diabetes mellitus	Polyphagia + weight loss; rapid bilateral cataracts in dogs	Fasting hyperglycemia + glucosuria + fructosamine
Diabetes insipidus (central/nephrogenic)	Extremely dilute urine regardless of hydration status	Water deprivation test +/- response to desmopressin
Hyperadrenocorticism	Pot-bellied appearance, symmetric truncal alopecia	LDDS or ACTH stimulation test
Chronic kidney disease	Isosthenuria with concurrent azotemia	Creatinine, SDMA, urine protein:creatinine ratio
Hypercalcemia (any cause)	Produces a nephrogenic-DI-like picture	Ionized calcium
Pyometra	Intact female, recent estrus, +/- vaginal discharge	Abdominal ultrasound, neutrophilia on CBC
Hyperthyroidism (cat)	Weight loss + polyphagia + tachycardia	Total T4
Psychogenic polydipsia	Young dog, otherwise normal labs	Water deprivation test — urine concentrates appropriately
Hepatic insufficiency / PSS	+/- post-prandial neurologic signs	Pre- and post-prandial bile acids
Hypoadrenocorticism (Addison's)	Hyperkalemia + hyponatremia	ACTH stimulation test

Pearl: Water deprivation testing differentiates psychogenic polydipsia from true diabetes insipidus, but only after every other cause on this list has been excluded first — it's risky in a dehydrated or azotemic patient.

2. Hypercalcemia — "GOSH DARN IT"

Cause	Key Clue
Granulomatous disease	History of fungal or chronic inflammatory disease
Osteolytic	Primary or metastatic bone lesion
Spurious	Lipemia or lab error — recheck ionized calcium
Hyperparathyroidism (primary)	Older dogs; low-normal phosphorus; parathyroid mass on cervical ultrasound
D — Vitamin D toxicosis	Cholecalciferol rodenticide, psoriasis cream, some houseplants — calcium AND phosphorus both elevated
Addison's disease	Concurrent hyperkalemia + hyponatremia
Renal disease	Uncommon cause — more often a consequence of hypercalcemia
Neoplasia	Lymphoma (most common malignant cause in dogs) or anal sac apocrine gland adenocarcinoma — check rectally
Idiopathic	Cats specifically; diagnosis of exclusion
Toxicosis	Same category as vitamin D toxicosis above

Pearl: Elevated calcium AND elevated phosphorus together points toward vitamin D toxicosis or renal failure; primary hyperparathyroidism classically shows low-normal phosphorus since PTH promotes phosphate excretion. Always palpate the anal sacs and peripheral lymph nodes before running an expensive workup — the two most common causes are found on physical exam.

3. Anemia Workup

Category	Subtype	Key Clue
Regenerative — hemolytic	IMHA	Spherocytes, autoagglutination, positive Coombs

Category	Subtype	Key Clue
		test
Regenerative — hemolytic	Heinz body hemolysis	Onion/garlic/zinc/acetaminophen exposure; Heinz bodies on smear
Regenerative — hemolytic	Hemotropic infection	Babesia, Mycoplasma haemofelis — organisms visible on blood smear
Regenerative — blood loss	External or internal hemorrhage	Low total protein (whole blood is lost, not just RBCs)
Non-regenerative	Anemia of chronic disease/inflammation	Mild, normocytic normochromic
Non-regenerative	Chronic kidney disease	Lack of erythropoietin production
Non-regenerative	Bone marrow disease	FeLV, myelophthisis, aplastic anemia

Pearl: Total protein is the fastest way to separate blood loss anemia (low TP — plasma proteins are lost along with red cells) from hemolytic anemia (normal/high TP — only red cells are being destroyed).

4. Pediatric Diarrhea (Puppies/Kittens)

Cause	Distinguishing Clue
Parvovirus	Unvaccinated puppy; hemorrhagic diarrhea + vomiting + neutropenia
Canine coronavirus	Milder GI signs, less systemic illness; worsens outcomes when co-infecting with parvo
Distemper (enteric phase)	Precedes or accompanies respiratory signs; neuro signs follow weeks later
Intestinal parasites	Roundworms (potbelly), hookworms (anemia), coccidia, Giardia — confirmed on fecal exam
Dietary indiscretion / diet change	Self-limiting, history-based, systemically well patient

Pearl: Neutropenia + hemorrhagic diarrhea + vomiting in an unvaccinated puppy is parvovirus until proven otherwise — but the fecal antigen ELISA can be falsely negative very early in disease or falsely positive shortly after a modified-live vaccine.

5. Icterus Localization

Location	Key Lab Pattern
Prehepatic (hemolysis)	Anemia present; normal-to-mild liver enzyme elevation; unconjugated bilirubinemia
Hepatic	Elevated ALT/AST; +/- low albumin, BUN, cholesterol (synthetic dysfunction)
Post-hepatic (obstructive)	Disproportionate ALP/GGT elevation; elevated cholesterol; +/- pale/acholic feces

Pearl: Check the PCV first. If the patient is anemic, think prehepatic/hemolytic before chasing a liver workup.

6. Tick-Borne Disease Overlap

Disease	Vector	Key Distinguishing Feature
Lyme disease	Ixodes tick	Shifting-leg polyarthritis; glomerulonephritis (Lyme nephritis) in chronic cases; Northeast/upper Midwest
Ehrlichiosis	Brown dog tick	Often severe thrombocytopenia, pancytopenia in chronic phase; more common South/Southwest
Anaplasmosis	Ixodes tick	Acute polyarthritis + thrombocytopenia; geographically overlaps with Lyme
Rocky Mountain Spotted Fever	Dermacentor tick	Vasculitis: petechiae, edema, CNS signs; can be peracute and fatal

Pearl: Thrombocytopenia is common to all four — differentiate by geography/vector and specific organ involvement (joints/kidney for Lyme, vasculitis/CNS for RMSF). Combination screening tests check Lyme, Ehrlichia, Anaplasma, and heartworm together.

7. Feline Lower Urinary Tract

Cause	Key Feature	Management Note
Feline idiopathic cystitis	Most common cause overall; diagnosis of exclusion; stress-associated	Environmental enrichment/stress reduction (MEMO)

Cause	Key Feature	Management Note
Struvite uroliths	Alkaline urine	Can potentially dissolve with a therapeutic urinary diet
Calcium oxalate uroliths	Acidic urine; more common in older cats	Cannot be dissolved medically — requires removal
Urethral plugs	Mix of struvite crystals + mucoprotein	Classic cause of male cat obstruction, distinct from true uroliths

Pearl: Struvite stones can be dissolved with diet; calcium oxalate stones cannot — this distinction is tested directly as a "what's your next step" question.

Equine Clusters

8. Neurologic Disease Cluster

Disease	Key Distinguishing Feature	Diagnostic Test
EPM	Asymmetric signs; opossum fecal exposure	CSF:serum antibody ratio
EHV-1 myeloencephalopathy	Often outbreak setting; may follow respiratory/abortion signs; urinary bladder paralysis common	PCR (nasal swab/blood)
West Nile virus	Mosquito-transmitted; muscle fasciculations, hyperesthesia	IgM capture ELISA
Rabies	Rapidly progressive, behavior change, ascending paralysis; always a differential given human exposure risk	No antemortem test — postmortem brain histopath/FA
Cervical stenotic myelopathy ("Wobblers")	Young, growing, large-breed horses; symmetric ataxia, worse in pelvic limbs	Cervical radiographs/myelography

Pearl: Rabies must always stay on the differential list for any horse with rapidly progressive, unexplained neurologic disease — use PPE and don't perform an oral exam or necropsy without precautions.

9. Adult Horse Diarrhea Cluster

Cause	Key Clue
Salmonellosis	Fever, severe illness; fecal culture needs multiple samples (intermittent shedding)
Potomac Horse Fever	Seasonal (summer/fall), near water/aquatic insect exposure; PCR for Neorickettsia risticii
NSAID toxicity / right dorsal colitis	History of NSAID (phenylbutazone) use; hypoproteinemia; thickened right dorsal colon on ultrasound
Sand enteropathy	Sandy soil/pasture exposure; sand settles out on a fecal float-in-water test
Antibiotic-associated colitis	Recent antibiotic use disrupting normal flora

Pearl: Always ask about recent NSAID or antibiotic use in an adult horse with colitis — iatrogenic causes are common and easy to overlook.

Bovine Clusters

10. Downer Cow Differential

Cause	Key Clue
Hypocalcemia (milk fever)	Recently calved; responds to IV calcium
Toxic (coliform) mastitis	Abnormal udder; systemic toxemia, cold extremities/shock
Calving-related nerve damage (obturator/peroneal)	History of dystocia/prolonged calving; normal calcium; classic "dog-sitting" posture
Hypomagnesemia (grass tetany)	Lactating cows on lush pasture; hyperesthesia/tetany precedes recumbency

Pearl: Response to IV calcium is the fastest way to separate milk fever from the other causes — if the cow doesn't respond, look for nerve damage or toxic mastitis instead.

11. Vesicular Disease

Disease	Key Feature
Foot-and-mouth disease	Foreign animal disease in the US; affects cattle/swine/sheep/goats; reportable
Vesicular stomatitis	Endemic in parts of the US; insect-transmitted (Culicoides/black flies); zoonotic

Pearl: FMD and vesicular stomatitis cannot be told apart on clinical signs alone — any vesicular lesion in cattle requires immediate reporting for laboratory differentiation, not a treatment plan.

Cross-Species: Toxicology & Regulatory

12. Rodenticide Toxicities

Toxin	Mechanism	Onset	Key Signs	Treatment
Anticoagulant (warfarin-type, brodifacoum)	Inhibits vitamin K epoxide reductase	Delayed 2–5 days	Bleeding; prolonged PT before PTT	Vitamin K1 (extended course for long-acting types)
Bromethalin	Neurotoxin causing cerebral edema	Hours to days	Tremors, seizures, hindlimb paresis	No antidote; mannitol/supportive care, early decontamination
Cholecalciferol (vitamin D3)	Hypercalcemia/hyperphosphatemia → soft tissue mineralization	Delayed 1–2 days	Renal failure, soft tissue mineralization	Prolonged/difficult (IV fluids/diuresis, calcitonin, bisphosphonates); poor prognosis

Pearl: "Rat poison" is not one toxidrome — it's three completely different mechanisms and treatments. Always ask to see the packaging/active ingredient before starting treatment; decontamination alone, without knowing which type, can miss the correct antidote window.

13. Reportable / Foreign Animal Disease Reporting

Examples: foot-and-mouth disease, African swine fever, highly pathogenic avian influenza, classical swine fever, screwworm.

Reporting chain: notify the state veterinarian and/or USDA APHIS immediately upon suspicion — do not wait for confirmatory lab testing before reporting.

Pearl: Recognizing that something is a foreign animal disease and knowing who to call is the tested skill — there is often no treatment, and the correct "next step" answer is regulatory, not clinical.

14. Calculations — Don't Skip Practice

Fluid rates, CRIs, and drug dose conversions aren't a knowledge gap for most students — they're a speed gap. Practice a handful under timed conditions this week so the arithmetic doesn't eat into time you need for reasoning through the case itself.

Appendix: Swine Enteric Disease Recap

Quick reference from the piglet diarrhea deep-dive — included for completeness, though porcine is only ~5% of the exam.

Piglet Diarrhea by Age & Agent

Pathogen	Typical Age	Stool Character	Necropsy Clue
E. coli (ETEC)	Birth–4 days	Profuse, watery, yellow-white, non-bloody	Fluid-distended intestine, minimal gross damage (secretory)
C. perfringens type C	Birth–3 days (peracute)	Bloody; sudden death without diarrhea common	Hemorrhagic, necrotic small intestine
Rotavirus	~1–3 weeks	Yellow/white, watery-to-pasty	Villous atrophy on histopath
Coccidia (Cystoisospora suis)	~7–21 days	Yellow, pasty-to-watery, +/- mucus	Occurs despite good sanitation — oocysts resist disinfectants
TGE / PEDV / PDCoV (coronaviruses)	Any age; near-100% mortality if <1 week old	Profuse watery diarrhea PLUS prominent vomiting	Severe villous atrophy

E. coli Fimbrial Antigens

Fimbria	Age Affected	Disease
F4 (K88)	Neonatal through post-weaning — only type whose receptor persists with age	Neonatal and post-weaning diarrhea
F5 (K99), F6 (987P), F41	Neonatal only — receptor disappears with age	Neonatal diarrhea
F18	Post-weaning only — receptor develops later	Post-weaning diarrhea, or edema disease if Shiga toxin (Stx2e)-producing

Pearl: *F18 + Shiga toxin (Stx2e) causes edema disease, which hits the best-doing, fastest-growing weaned pigs hardest (they eat the most toxin-producing bacteria) and presents with vascular damage — subcutaneous edema, ataxia, sudden death — rather than typical secretory diarrhea.*

Coronaviral Enteritis: Key Points

Vomiting plus profuse watery diarrhea distinguishes TGE/PEDV/PDCoV from the bacterial and rotavirus causes above. Mortality approaches 100% in neonates under about 1 week old but drops sharply with age. In a naive herd, disease sweeps through all ages at once; in an endemic herd, colostral immunity confines disease mostly to piglets from non-immune litters. PRCV, a naturally occurring TGEV deletion mutant, causes mild respiratory disease but provides partial cross-protection against TGE.